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RESEARCH ARTICLE

Improved Artificial Potential Field Method for UAV Path Planning

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ABSTRACT Path planning for uncrewed aerial vehicles (UAVs) is a core technology for autonomous navigation. The artificial potential field (APF) method, known for its computational efficiency and strong real-time performance, has been widely applied. However, the traditional approach suffers from three inherent defects: target unreachability, local minima, and local path oscillation. To address these issues, this paper proposes an Improved Artificial Potential Field (IAPF) method, innovatively incorporating three types of optimization mechanisms: 1) A target point distance-weighted function to reconstruct the repulsive field, ensuring terminal convergence; 2) A local exploration factor to achieve escape from local minima; 3) A directional weighting factor and a forward-looking decision-making mechanism to suppress path oscillations. Moreover, experiments have demonstrated the ability of the proposed algorithm to address the issues in traditional APF, as well as its robustness and superiority in complex three-dimensional environments.

INDEX TERMS Path planning, improved artificial potential field, target unreachability, local minima, local path oscillation.

I. INTRODUCTION

In recent years, UAVs have been widely used in both military and civilian fields such as military air combat [1] and precision agriculture [2] due to their excellent mobility and flexibility. During flight, UAVs face threats from obstacles in the environment. With the popularization of measurement and control sensing, machine vision, and other technologies, UAV autonomous flight technology has gradually matured. Among these technologies, path planning, as a key component of UAV autonomous flight, determines whether a UAV can safely complete its mission [3]. Path planning algorithms can be broadly classified as either traditional or intelligent approaches. Traditional path planning algorithms include the A* algorithm [4], Dijkstra algorithm [5], APF method [6], Rapidly-exploring Random Tree (RRT) [7], and Rapidly-exploring Random Tree Star (RRT*) [8], while intelligent path planning algorithms include Particle Swarm Optimization [9], Ant Colony Optimization [10], and Simulated Annealing [11].

Although algorithms such as the A* and RRT algorithms can plan the shortest path, their computational time increases exponentially in complex scenarios, resulting in poor real-time performance [12], [13]. Intelligent algorithms like Particle Swarm Optimization and Ant Colony Optimization can reduce computational time but suffer from low search efficiency and difficult parameter tuning [14], [15].

The APF method, proposed by Khatib in 1985, is a local path planning algorithm based on virtual force fields. Widespread adoption in obstacle avoidance and trajectory optimization has been achieved by this approach, owing to its rapid computational convergence and deterministic temporal performance [16], [17]. However, it also suffers from issues such as target unreachability, local minima, and local path oscillation. Scholars worldwide have conducted extensive research on the APF method. To address the local minima

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problem, Min and Nam [17] introduced a dynamic adjustment mechanism based on the estimated time of collision to correct the path direction in real time within the potential field, thereby preventing drones from falling into local minima. Sang et al. [18] proposed an adaptive inertia weight based on the optimal success rate strategy to improve the APF and introduces a chaos strategy to avoid local minima. Lin et al. [19] and Lin et al. [20] modified the repulsive potential field to make it always perpendicular to the flight direction to solve the local minima problem. Xu et al. [21] introduced angular and velocity adjustment factors and proposes an auxiliary obstacle avoidance force to achieve early obstacle avoidance and prevent drones from falling into local minima. Lv et al. [22] introduced virtual sub-goals to guide robots to avoid stagnation points and overcome the local minima problem.

In addition, for solving the target unreachability problem, Yang and Su [23] addressed the issue of motion stagnation caused by force balance by introducing a dynamic repulsion gain parameter. Zhang et al. [24] solved the target unreachability problem by adjusting the distance threshold and the regulation factor of the gravitational field function. For solving the local path oscillation problem, Deng et al. [25] a globally reliable path is generated by the bidirectional A* algorithm, ensuring the correctness of the navigation mission at the strategic level. Xi et al. [26] proposed a hierarchical framework combining APF and deep reinforcement learning (DRL) to solve local path oscillation by incorporating the velocity information of the target. Hao et al. [27] enabled drones to plan more rational and smoother paths in complex environments by introducing collision risk assessment, virtual sub-targets, and deflection angle suppressors.

To simultaneously address the aforementioned issues in traditional APF for path planning, this paper proposes an IAPF method. First, obstacles in the environment are classified into rigid and non-rigid obstacles, and different threat distances are set for different types of obstacles to construct the obstacle model of the flight environment. Then, the IAPF is used for path planning. The target unreachability problem is solved by introducing a target point distance-weighted function. The local minima problem is addressed by introducing a local exploration factor to guide the UAV to jump out of local minima. The local path oscillation problem is mitigated by introducing a directional weighting factor and a forward-looking decision-making mechanism. Finally, the generated path is smoothed. The main contributions of this paper are: systematically solving the three inherent classic problems of APF and integrating them into a unified, high-performance algorithm framework; further enhancing the practicality and usability of the generated path; comprehensively and deeply verifying the superiority of the algorithm through experiments; and providing an efficient, low-computational-power-demand solution for path planning in complex environments.

II. MATERIALS AND METHODS

A. THE APF METHOD

The APF method guides a UAV by simulating forces. It generates an attractive force pulling the UAV toward the target goal and repulsive forces pushing it away from surrounding obstacles. Autonomous navigation emerges from the UAV's trajectory following the negative gradient of this synthesized potential landscape, where the convex combination of attraction and repulsion forces guarantees collision-free path generation through differential vector superposition.

The gravitational and repulsive potential field functions can be expressed as follows:

$$U_{att}(X) = ck_{att}d(X, X_e)^2 \tag{1}$$

$$U_{rep}(X) = \begin{cases} ck_{rep}(\frac{1}{d(X, X_t)} - \frac{1}{\rho})^2, d(X, X_t) \leq \rho \\ 0, d(X, X_t) > \rho \end{cases} \tag{2}$$

where, c represents the constant term. k_{att} and k_{rep} denote the gravitational and repulsive coefficients, respectively. $X = (x, y, z)$ indicates the current position of the UAV, while $X_e = (x_e, y_e, z_e)$ and $X_t = (x_t, y_t, z_t)$ represent the locations of the target point and the obstacle, respectively, $d(X, X_e) = \sqrt{(X_e - X)^2}$ and $d(X, X_t) = \sqrt{(X_t - X)^2}$ represent the Euclidean distance from the UAV to the target point and the obstacle, and ρ signifies the obstacle threat distance.

The gravitational force $\boldsymbol{F}_{\text{att}}(\boldsymbol{X})$ and repulsive force $F_{rep}(X)$ can be obtained by finding the negative gradient of the gravitational and repulsive potential fields, which is expressed as follows:

$$F_{att}(X) = -\nabla(U_{att}) = -2ck_{att}d(X, X_e)\frac{\partial d(X, X_e)}{\partial X} \tag{3}$$

$$F_{rep}(X) = -\nabla(U_{rep}) = \begin{cases} 2ck_{rep}(\frac{1}{d(X, X_t)} - \frac{1}{\rho})\frac{1}{d(X, X_t)^2}\frac{\partial d(X, X_t)}{\partial X}, d(X, X_t) \leq \rho \\ 0, d(X, X_t) > \rho \end{cases} \tag{4}$$

The UAV experiences a resultant force, which stems from multiple repulsive forces and an attractive force. This resultant force dictates the UAV's subsequent flight heading, enabling it to navigate safely to the target while circumventing obstacles. The formula is presented below:

$$F_{all}(X) = F_{att}(X) + \sum_{i=1}^{n} F_{rep\_i}(X) \tag{5}$$

The APF method is straightforward to implement. However, a significant limitation arises because the resultant force is a simple vector sum of the attractive and repulsive forces. This leads to three inherent disadvantages:

(1) As the UAV draws near the target point, the target - generated gravitational potential field diminishes, while the obstacle - generated repulsive potential field grows stronger with the UAV's approach. Therefore, in the path planning

process, if there is an obstacle near the target point, the repulsive force may exceed the gravitational force, resulting in the combined force pointing away from the direction of the target point, causing the UAV to fall into the target unreachability problem as shown in Figure 1.

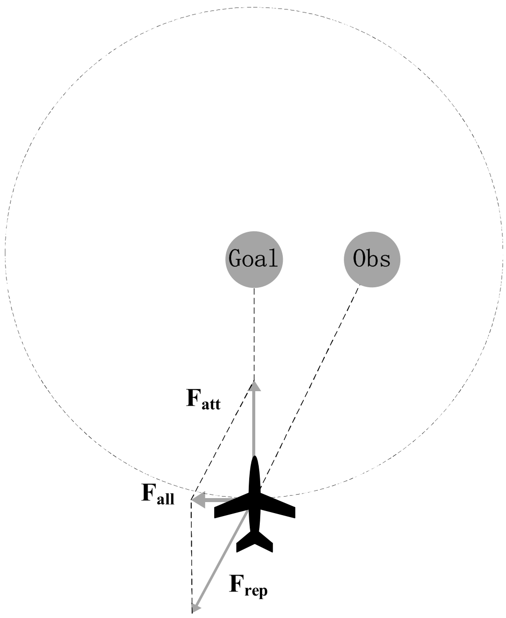


FIGURE 1. The problem of target unreachability.

(2) As the UAV moves in the direction of the falling potential field during path planning, it can only make decisions based on the local information of the current position and cannot obtain global information. In complex scenarios, this local information can lead to a situation where the combined force is zero and the UAV is unable to choose a better path, thus falling into a local minima trap. As shown in Figure 2.

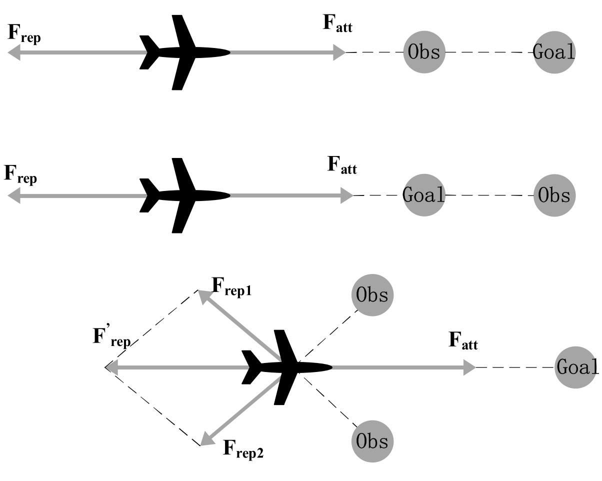


FIGURE 2. The problem of local minima.

(3) When the UAV approaches the target point, the gravitational potential field plays a dominant role, causing the UAV to fly towards the target point; when the UAV encounters an obstacle, the repulsive potential field dominates, causing the UAV to move away from the obstacle point. Due to the interaction of these two forces, the UAV may experience repeated gravitational and repulsive changes in the vicinity of the obstacle, which may result in the UAV continuously approaching and moving away from the obstacle, forming a back-and-forth oscillatory motion pattern and thus creating a local path oscillation problem, as shown in Figure 3.

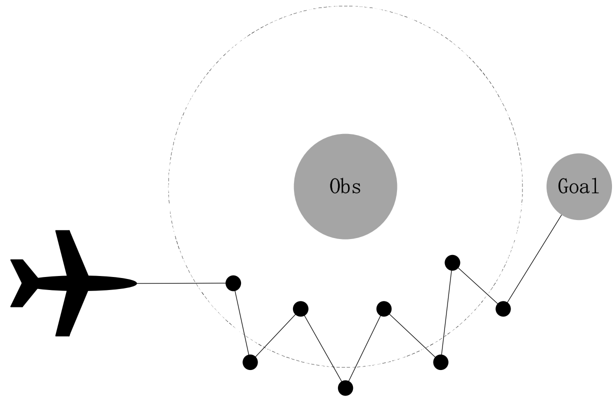


FIGURE 3. The problem of local path oscillation.

B. THE IAPF METHOD

To address the problems of the above APF method, this paper performs path planning by improving the algorithm and combining the ideas of environment awareness and deep learning. The process is shown in Figure 4.

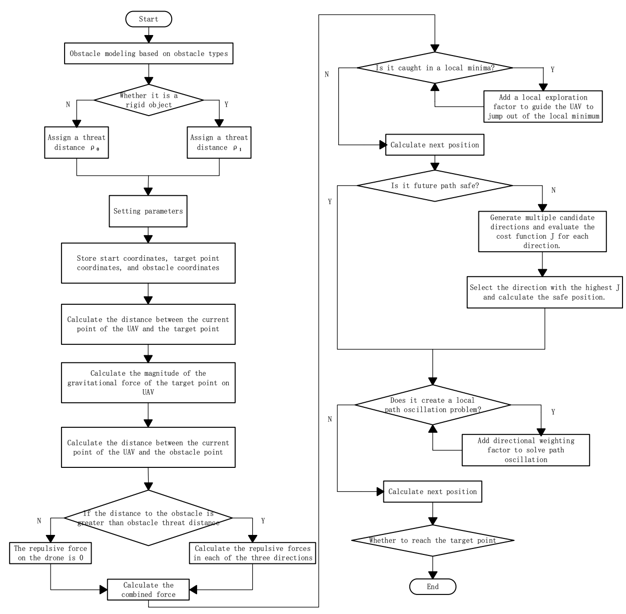


FIGURE 4. Flow chart of IAPF method path planning.

C. IMPROVEMENT OF TARGET ATTAINABILITY

To tackle the target unreachability issue, this paper enhances the potential field function. This ensures that even when the target is hard to reach, the gravitational force from the target point can still guide the UAV to the target position. In this improvement, the gravitational potential field function remains as per the traditional method. However, a target point distance weighting function $\varphi(X, X_e) = -\frac{e^{d(X,X_e)}}{1+e^{d(X,X_e)}} + \frac{1}{2}$ is added to the repulsive potential field function. The enhanced repulsive potential field function is defined as follows:

$$U_{rep}(X) = \begin{cases} (\frac{1}{2} - \frac{1}{1+e^{-d(X,X_e)}})ck_{rep}(\frac{1}{d(X,X_t)} - \frac{1}{\rho})^2, d(X,X_t) \leq \rho \\ 0, d(X,X_t) > \rho \end{cases} \tag{6}$$

where, c is a constant term and n is a positive number. The new repulsive function is derived by computing the negative gradient of the enhanced repulsive potential field function, as expressed below:

$$F_{rep}(X) = -\nabla(U_{rep}) = \begin{cases} F_{rep1} + F_{rep2}, d(X,X_t) \leq \rho \\ 0, d(X,X_t) > \rho \end{cases} \tag{7}$$

where, F_{rep1} is the repulsive force of the obstacle facing the UAV and F_{rep2} is the gravitational force of the target point facing the UAV, defined as follows:

$$F_{rep1}(X) = 2ck_{rep}(\frac{1}{d(X,X_t)} - \frac{1}{\rho})\frac{1}{d(X,X_t)^2}(\frac{1}{2} - \frac{1}{1+e^{-d(X,X_e)}})\frac{\partial d(X,X_t)}{\partial X} \tag{8}$$

$$F_{rep2}(X) = ck_{rep}(\frac{1}{d(X,X_t)} \frac{1}{\rho})^2 e^{-d(X,X_e)}\frac{1}{(1+e^{-d(X,X_e)})^2}\frac{\partial d(X,X_e)}{\partial X} \tag{9}$$

As the UAV approaches the target point, $d(X, X_e)$ diminishes and F_{rep1} nears 0. At this juncture, the UAV's repulsive force is primarily determined by F_{rep2}, F_{rep1}, and F_{rep2}, which collectively dictate the new repulsive force's magnitude and direction, as illustrated in Figure 5.

Through the above steps, when the UAV gradually approaches the target point, causing the gravitational force of the target point on the UAV to gradually decrease, thus causing the target unreachability problem, by introducing a target point distance weighting function, the balance of the combined forces can be effectively adjusted as the UAV approaches the target point, gradually weakening the influence of the repulsive force as the distance between the UAV and the target point gradually decreases, while enhancing the effect of the gravitational force to ensure that the UAV can continue to be attracted by the gravitational force when encountering obstacles, while reducing the influence of the repulsive force and continuing to guide the UAV closer to the target point.

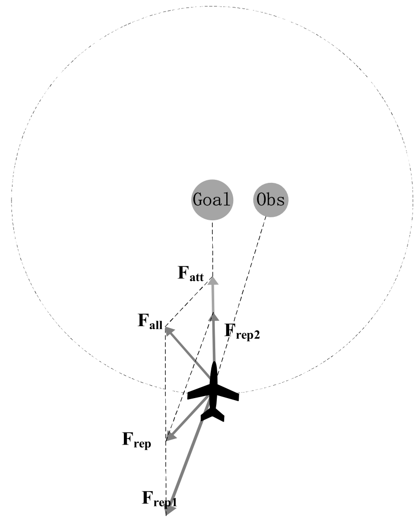


FIGURE 5. Improved ensemble force model.

D. IMPROVEMENT OF LOCAL MINIMA PROBLEMS

For the local minima problem, the UAV is guided out of the local minima trap by introducing a local exploration factor of ε. The main steps are as follows:

Step 1: Assume the potential field function is $U(X)$. Calculate the gradients of the potential field function $U(X)$ at the current position $G = [G(x), G(y), G(z)]$ in three directions: $G(x) = \frac{\partial U(X)}{\partial x}$, $G(y) = \frac{\partial U(X)}{\partial y}$, and $G(z) = \frac{\partial U(X)}{\partial z}$, as well as the gradient vector.

Step 2: When the magnitude of the gradient $|G|$is less than a very small threshold ε_g (e.g., 0.001 m), and the distance between the current position and the target point is greater than another threshold ε_d(e.g., 0.5 m), the UAV is determined to be trapped in a local minimum.

Step 3: Introduce the local exploration factor ε and add it to the corresponding direction of the current position $\boldsymbol{X}$ to get the new position $X_{new} = X + \varepsilon$.

Step 4: Recalculate the gradient $G(X_{new})$ at the new position X_{new}, and if $G(X_{new}) \neq 0$, the local minima has been successfully jumped out.

Step 5: Repeat the above steps if you run into the local minima problem until you reach the target point.

This procedure allows the UAV to escape local minima. The algorithm first identifies the specific direction in which the UAV's motion is trapped within a local potential field. By then introducing a local exploration factor, it adds a stochastic perturbation to the current motion direction. This enables the UAV to explore alternative paths and escape the local minimum.

E. IMPROVEMENT OF LOCAL PATH OSCILLATION PROBLEMS

To address the local path oscillation issue, this paper presents an optimization approach that uses steering angle control and incorporates a directional weighting factor. This method effectively restrains the steering angle, thereby reducing

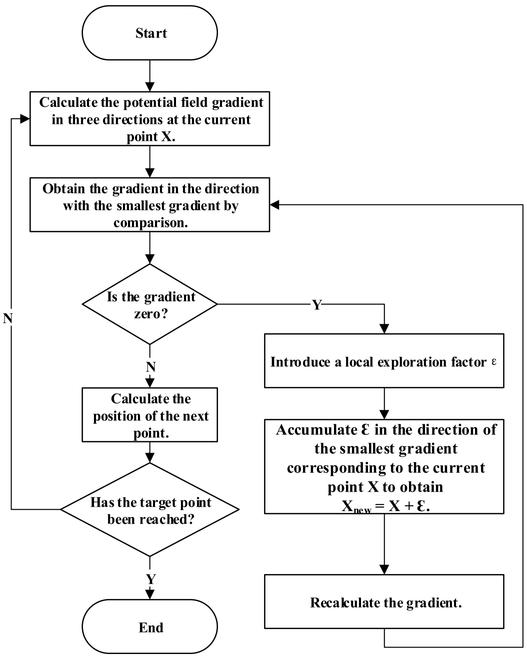


FIGURE 6. Flow chart of local minima problem optimization.

oscillation when the UAV makes sharp turns between adjacent path points.

To calculate the resultant force F_{n+1} acting on the UAV at position X_n as it moves toward the theoretical next path point X_{n+1}, the paper introduces an optimization method. This method also factors in the resultant force F_n from the preceding path point X_{n-1} to X_n. By determining the angle change $\Delta\alpha$ between F_n and F_{n+1}, the UAV can adjust its trajectory. The actual next path point X_{n+1} is then decided based on the Equation (10).

$$X_{n+1} = \begin{cases} X_n + L \cdot (m_1 \cdot w_n + m_1' \cdot w_{n+1}), \Delta\alpha \le \theta \\ X_n + L \cdot (m_2 \cdot w_n + m_2' \cdot w_{n+1}), \Delta\alpha > \theta \end{cases} \tag{10}$$

where, $m_i$and m_i' are the directional weight factors, w_n and w_{n+1} are the directional vectors of F_n and F_{n+1}, respectively, L is the motion step, and θ is the angular threshold. The values of the directional weighting factor and the angular threshold can be referred to as follows:

$$\begin{cases} m_i = m_i', & if \quad 0 \le \Delta\alpha \le \frac{\pi}{6} \\ m_i > m_i', & if \quad \frac{\pi}{6} < \Delta\alpha \le \frac{\pi}{3} \\ m_i \gg m_i', & if \quad \Delta\alpha > \frac{\pi}{3} \end{cases} \tag{11}$$

Through the above steps, when the angle of the UAV deviates from the current heading is small, the deviation can be considered as a small area of fine adjustment, and the UAV can continue to fly in the preset direction; when the angle of the UAV deviates from the current heading is large, the path is prone to oscillation, and by adding the direction weighting factor, the heading will be as close as possible to the initial heading to avoid too large a turning angle, thus suppressing the oscillation. As depicted in Figure 7, X_{n-1} and X_n represent the UAV's preceding and current path point positions, respectively, while X_{n+1} denotes the theoretical subsequent path point. Corresponding theoretical forces include the gravitational force F_{att}, repulsive force F_{rep}, and their vector resultant F_{all}. The actualized next waypoint X'_{n+1} is determined through force recalibration, incorporating adjusted gravitational F'_{att}, repulsive F'_{rep}, and resultant F'_{all} forces, where directional continuity constraints ensure smooth transitions between consecutive path segments.

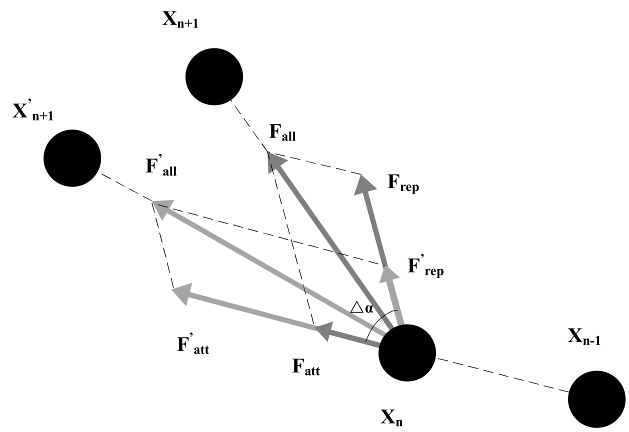


FIGURE 7. Calculation of the force diagram for the next position of UAV.

In addition, this paper introduces a forward-looking decision-making mechanism into the algorithm. This mechanism, based on the model predictive control framework, predicts the future path states for multiple steps ahead at each decision point and evaluates the safety and goal orientation of each candidate direction through a multi-objective cost function. Specifically, the cost function takes into account both the directional consistency term and the safety indicator term, and ultimately selects the direction with the optimal comprehensive score for execution. This approach transforms the pure force-driven method into a predictive optimization decision, significantly improving the success rate and quality of path planning. The predictive model and cost function are shown in Equations (12) and (13).

$$X_{k+i}X_k + v \cdot \vec{u}, i = 1, 2, \ldots, N \tag{12}$$

$$J(\alpha) = \beta_1 \cdot C_{dir}(\alpha) + \beta_2 \cdot C_{safe}(\alpha) \tag{13}$$

where, v represents the current velocity of the UAV, u is the predicted direction vector, and β_1 and β_2 are the weighting coefficients.

F. PATH OPTIMIZATION

When generating a UAV's flight path, the most basic requirements are to ensure safety and to keep the range as short as possible. The drone needs to avoid obstacles to ensure safety and to minimize the flight distance given physical constraints such as power. However, to ensure safety, the UAV will move away from obstacles, resulting in an increased flight distance, while to minimize range, the UAV may ignore the threat of obstacles, reducing safety. To address this issue, different threat distances can be assigned to different types of obstacles by adjusting the parameters in the APF method, the UAV can get as close as possible to the less threatening obstacles while avoiding the more threatening ones, thus balancing the requirements of UAV flight safety and range minimization. This path optimization method takes into account the physical constraints of the UAV and the threat of obstacles, and generates a more reasonable and feasible trajectory.

After the overall path is generated, there may be some path points with excessively steep turning angles. Since UAVs cannot turn quickly during actual flight, this may lead to flight safety issues. In this paper, B-spline fitting is adopted to smooth the path. By using piecewise low-order polynomials under the constraint of overall continuity and differentiability, the method approximates the given control points as closely as possible to achieve curve smoothing. Through the above methods, the path can be smoothed, and the generated path better meets the flight requirements of UAVs.

III. RESULTS

A. EXPERIMENTAL SIMULATION AND COMPARISON

This section verifies the proposed IAPF method through simulations. First, simulations and comparisons are conducted to validate the solution to the target unreachability problem by introducing a target point distance-weighted function. Second, simulations are performed to verify the effectiveness of the local exploration factor in guiding the UAV to escape from local minima. Third, simulations and comparisons are carried out to validate the solution to the local path oscillation problem by introducing a directional weighting factor and a forward-looking decision-making mechanism. Finally, the robustness and performance improvement of the algorithm are verified through ablation and comparative experiments in complex environments. The planning area is set to 20 m × 20 m × 20 m and 30 m × 30 m × 30 m. Rigid obstacles are configured as cylindrical fillers, with repulsive forces originating from the nearest point on the obstacle surface to the UAV. Non-rigid obstacles are configured as spherical fillers, with threat distances of $\rho = \rho_0$ and $\rho = \rho_1$. All experiments were conducted on an Intel Core i5-12400F with 16GB RAM using MATLAB R2023b.

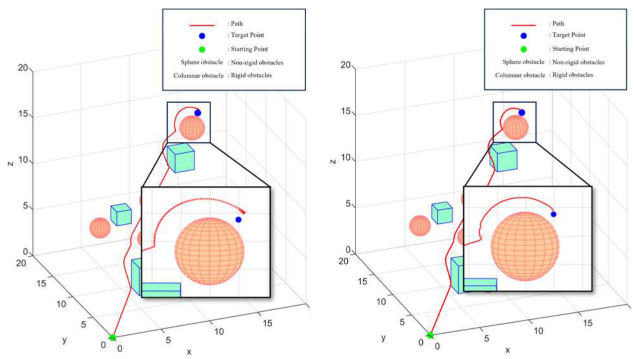
B. SIMULATION VERIFICATION OF IMPROVED TARGET UNREACHABILITY PROBLEM

In this paper, we solve the target unreachability problem by introducing the target point distance weighting function, and compare the path planning situation between the APF method and the method of this paper in the same obstacle environment.

The parameter settings are given in Table 1. There are 5 rigid obstacles and 5 non-rigid obstacles included in the track planning area, the center points of the rigid obstacles are [5,4,4], [5,10,8], [10,6,7], [10,10,5], [12,12,12], the length and width are [2,2,3], [1. 5,1.5,1.5], [3,3,5.5], [1,1,1], [1,1,1], [2,2,2]; the center point of the non-rigid obstacles is [2,2,2]; the coordinates of the center points of the non-rigid obstacles are [2,8,8], [6,6,7], [7,3,3], [8,10,9], [14,14,14] with radii 1, 0. 9, 0.8, 1, 1.2, respectively. Figure 8 shows the comparison of the two algorithms, the traditional algorithm produces the target unreachability problem, the trajectory will keep jittering when there is an obstacle near the target point and cannot reach the target point, while the improved algorithm can successfully reach the target point, which indicates that the improved algorithm can solve the target unreachability problem.

TABLE 1. Parameter description of target unreachability problem.

Parameters	Numerical value
Rigid obstacle threat distance/m	2
Non-rigid obstacle threat distance/m	2
Gravitational coefficient	15
Repulsion coefficient	10
Moving step/m	0.1
Starting point coordinates	[0,0,0]
Target point coordinates	[15,15,15]



(A) APF method (B) IAPF method

FIGURE 8. Comparison of target unreachability problem.

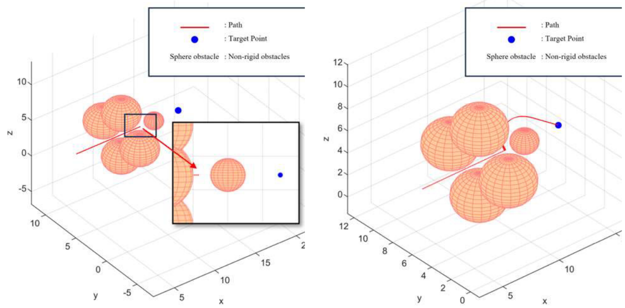
C. SIMULATION VERIFICATION OF IMPROVED LOCAL MINIMA PROBLEM

In this paper, the local exploration factor is introduced to guide the UAV to jump out of the local minima, and the path planning of the APF method is compared with the method in this paper in the same obstacle environment.

The parameter settings are given in Table 2. There are five non-rigid obstacles in the path planning region, and the coordinates of the center points of the non-rigid obstacles are [8,8,5], [8,2,5], [8,5,8], [8,5,2] and [12,5,5] with radii of 2, 2, 2, 2, 2 and 1. Figure 9 shows the comparison of the two algorithms, and the traditional algorithm produces a local minima problem at the point [10,5,5], which is in the x, y, z The combined force in the direction of [360,0,0], while the improved algorithm successfully jumps out of the local minima to reach the target point, which shows that the improved algorithm can solve the local minima problem.

TABLE 2. Parameter description of local minima problem.

Parameters	Numerical value
Rigid obstacle threat distance/m	2
Non-rigid obstacle threat distance/m	2
Gravitational coefficient	15
Repulsion coefficient	10
Moving step/m	0.1
Starting point coordinates	[0,5,5]
Target point coordinates	[15,5,5]

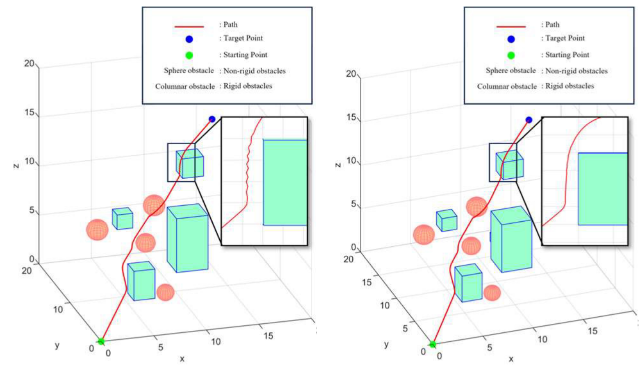


(**A**) APF method (**B**) IAPF method

FIGURE 9. Comparison of local minima problem.

TABLE 3. Parameter description of local path oscillation problem.

Parameters	Numerical value
Rigid obstacle threat distance/m	2
Non-rigid obstacle threat distance/m	2
Gravitational coefficient	15
Repulsion coefficient	10
Moving step/m	0.1
Starting point coordinates	[0,5,5]
Target point coordinates	[15,5,5]
Look ahead distance/m	2.0
Max look ahead steps	5
Number of random directions	10
Safety term bonus	1.0



(**A**) APF method (**B**) IAPF method

FIGURE 10. Comparison of local path oscillation problem.

D. SIMULATION VERIFICATION OF IMPROVED LOCAL PATH OSCILLATION PROBLEM

In this paper, the local path oscillation problem is solved by introducing the combined force angle change and direction weight factor, and the path planning situation between the APF method and the method of this paper in the same obstacle environment is compared.

The parameter settings are given in Table 3. There are 5 rigid obstacles and 5 non-rigid obstacles in the path planning area, and the center points of the rigid obstacles are [5,4,4], [5,10,8], [10,6,7], [10,10,5], [12,12,12], and the lengths and widths are [2,2,3], [1. 5,1.5,1.5], [3,3,5.5], [1,1,1], [1,1,1 [2,2,2]]; the coordinates of the center points of the non-rigid obstacles are [2,8,8], [6,6,7], [7,3,3], [8,10,9], [14,14,14] with radii of 1, 0.9, 0. 8, 1, 1.2, respectively. Figure 10 shows the comparison of the two algorithms, the traditional algorithm has the problem of local path oscillation, where the path points near the obstacle are constantly oscillating, while the improved algorithm obviously improves the local path oscillation problem, which shows that the improved algorithm can solve the local path oscillation problem.

E. ABLATION STUDY

This paper conducts UAV path planning using the APF method, and then successively incorporates improvements for the target unreachability problem, the local minima problem, the path oscillation problem, and B-spline fitting, comparing each improvement with the previous version.

The parameter settings are shown in Table 6. The simulation scenario is a 30 m × 30 m × 30 m airspace, with a total of 45 static obstacles, including 21 rigid cuboids (height 12–25 m, base dimensions 1.5–3.2 m) and 24 non-rigid spheres (radius 1.0–3.0 m). The obstacles are spatially distributed to cover the takeoff area, cruise segment, and the area near the target, with a minimum spacing of 0.5 m (rigid) and 0.9 m (non-rigid). The specific obstacle parameters are shown in Tables 7 and 8. In the traditional APF, improvements for target unreachability, local minima, path oscillation, and B-spline optimization are successively incorporated, and the paths before and after each improvement are compared in the same figure window. The experimental results are shown in Figure 11 and Figure 12.

In Figure 11(a), the path length is reduced from 128.2 m to 64.2 m, a decrease of 49.9%. This is mainly due to the target point distance-weighted function, which weakens the repulsive force as the UAV approaches the target, avoiding path elongation or target unreachability caused by excessive repulsive force. However, the increased computational load results in longer planning time. From Figure 11(a) to Figure 11(d), it can be seen that the path gradually becomes

TABLE 4. Parameter description of the ablation experiments.

Parameters	Numerical value
Rigid obstacle threat distance/m	2
Non-rigid obstacle threat distance/m	2
Gravitational coefficient	15
Repulsion coefficient	10
Moving step/m	0.2
Starting point coordinates	[0,5,5]
Target point coordinates	[30,30,20]
Look ahead distance/m	2.0
Max look ahead steps	5
Number of random directions	10
Safety term bonus	1.0

TABLE 5. Parameters of non-rigid obstacles.

x,y,z,r	5,5,5,1.5	10,8,12,1.2	14,13,9,1.8	20,18,11,1.0	25,22,18,1.4	8,20,10,1.6
	17,8,15,1.6	23,25,12,1.8	27,12,22,1.0	3,25,18,1.5	9,10,28,1.3	15,28,10,1.7
	12,25,18,1.3	18,5,22,1.7	22,15,8,1.1	28,8,20,1.9	7,15,25,1.4	13,22,5,1.2
	22,18,14,1.2	26,22,8,1.6	10,10,12,2.0	28,28,18,3.0	14,9,18,1.5	17,13,16,1.2

TABLE 6. Parameters of rigid obstacles.

x,y,z	8,12,11.25	12,18,7.75	16,5,10	20,22,9.75	24,8,12	28,15,8.25	6,25,10.5
	10,10,2.5	14,20,11.75	18,2,7.5	22,15,2.5	26,5,9.25	4,18,9.5	8,8,11
	12,22,2.5	16,12,12.25	20,28,8.5	24,10,10.25	28,20,10.75	16,16,10	13,8,6
L,W,H	1.8,2.5,22.5	3.2,1.8,15.5	2,2.2,20	2.8,1.5,19.5	1.5,2.8,24	2.2,2,16.5	3.0,1.8,21
	1.8,3,17.5	2.5,2.2,23.5	2,2.5,15	2.8,1.5,25	1.5,2.8,18.5	3.2,2,19	2.2,1.8,22
	1.8,2.2,16	2.5,1.5,24.5	3,2,17	2.2,2.8,20.5	1.8,3,21.5	2,2,20	2,2,12

smoother with the incorporation of improvements. Especially after smoothing, the path better conforms to the UAV's flight trajectory. This is because B-spline fitting eliminates the jagged path caused by abrupt changes in the force field in traditional APF while maintaining path continuity and differentiability. Meanwhile, in Figure 11(c), the planning time is reduced from 0.36 s to 0.12 s. This is because the forward-looking decision-making mechanism enables the UAV to detect obstacles in advance. Although the computational load per iteration increases, the overall number of iterations and path length are often reduced by avoiding collisions and oscillations, thereby improving overall planning efficiency.

Figure 12 illustrates the enhanced smoothness of the proposed algorithm. The traditional APF has an average curvature of 1.4028 and a maximum curvature of 3.1012, while the IAPF-generated path, after smoothing, has an average curvature of 0.3803 and a maximum curvature of 2.3100. The distribution of the x-coordinates of the path points also

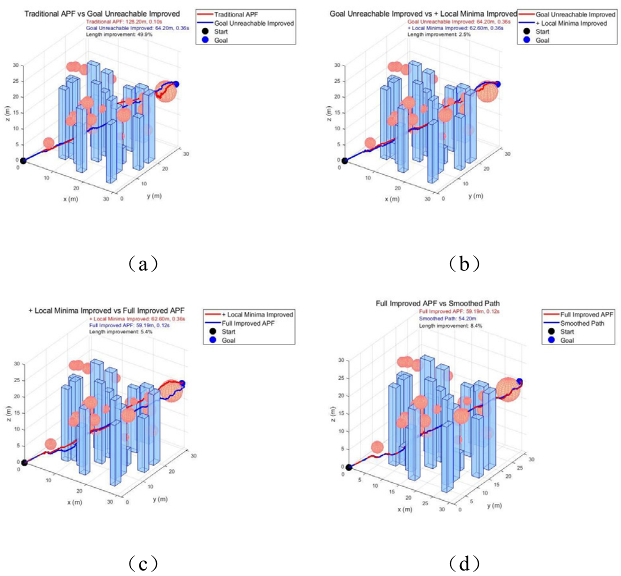


FIGURE 11. Comparison of IAPF improvements before and after.

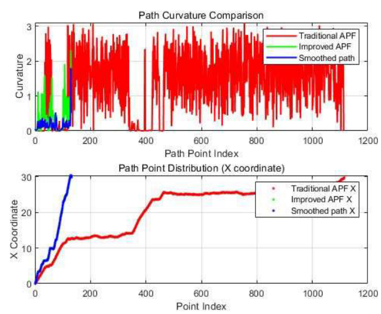


FIGURE 12. Path curvature display.

clearly shows that the traditional APF-generated path has multiple local back-and-forths, whereas the smoothed IAPF path has no local back-and-forths. Overall, the above results demonstrate the robustness and performance improvement of the IAPF algorithm.

F. MONTE CARLO SIMULATION EXPERIMENT

To further verify the robustness and convergence of the algorithm, this paper conducted 100 path planning experiments with random start and goal points in the same environment as the previous experiments. The experimental results successfully demonstrated the robustness and convergence of the proposed algorithm. The specific parameters and experimental results are shown in Table 7 and Table 8.

G. COMPARATIVE EXPERIMENTS

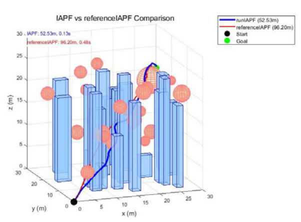
To further verify the algorithm's performance, this paper compares the paths generated by IAPF with those generated by RRT*, A*, and the IAPF algorithm in [27] under the parameters and obstacle environment listed in Table 4. The results are shown in Figure 13.

TABLE 7. Parameter settings for the monte carlo simulation experiment.

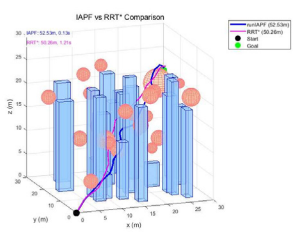
Parameters	Numerical value
Rigid obstacle threat distance/m	2
Non-rigid obstacle threat distance/m	2
Gravitational coefficient	1.5
Repulsion coefficient	2.5
Moving step/m	0.2
Target point threshold	0.4
Look ahead distance/m	2.0
Max look ahead steps	5
Number of random directions	10
Safety term bonus	1.0

TABLE 8. The result of Monte Carlo simulation experiment.

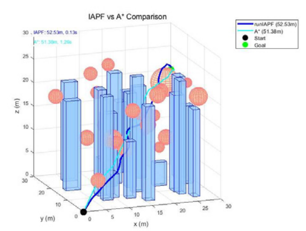
Algorithm	Success rate	Average endpoint error (±std) /m	Average curvature/m^{-1}
Traditional APF	80%	0.3020 (±0.0575)	1.2356
Target unreachability problem improved	97%	0.3009 (±0.0569)	0.7596
+ Improvement for local minima	97%	0.3123 (±0.0551)	0.7092
IAPF	100%	0.1867 (±0.0963)	0.3174



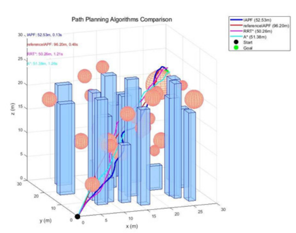
(a) IAPF vs IAPF in [27]



(b) IAPF vs RRT*



(c) IAPF vs A*



(d) IAPF vs Three Algorithms

FIGURE 13. Comparison of IAPF with other three algorithms.

As shown in Figure (a), the path planned by the proposed algorithm (blue) is smoother and shorter than that of the algorithm in [27] (red). The red parh exhibits noticeable oscillations and detours in the densely obstructed areas due to its failure to effectively address the path oscillation problem of traditional APF. The proposed algorithm, by introducing a forward-looking decision-making mechanism and directional weighting factor, achieves early obstacle avoidance and smooth turning, effectively suppressing oscillations and generating a path closer to the global optimum.

Figure (b) shows that although RRT* is a global planner that can theoretically converge to the optimal path, the path it generates in this experimental environment (purple) contains many unnecessary detours and zigzags, resulting in a significantly longer path length. This is because RRT* requires a large number of sampling points to find a high-quality path in complex environments, which is computationally expensive. In contrast, the proposed IAPF algorithm generates a direct, smooth, and shorter path (blue), demonstrating that under known environment maps, the proposed algorithm, which is reactive but augmented with predictive and optimization mechanisms, can outperform classical sampling-based global planning algorithms in terms of path quality.

In Figure (c), the A* algorithm searches in a discrete grid space to ensure that the shortest path is found, but its result (cyan) is a zigzag path composed of line segments, which does not conform to the UAV's dynamic constraints and must be post-processed for smoothing before flight. The proposed IAPF algorithm, on the other hand, plans directly in continuous space, generating a naturally smooth and flyable path without the need for post-processing. Although the path length may be slightly longer than the theoretical shortest path of A*, its actual flight efficiency is higher.

The experimental results show that the proposed IAPF algorithm successfully integrates the real-time performance of the APF method with the optimality of global planning algorithms. It not only addresses the inherent drawbacks of traditional APF but also outperforms classical algorithms such as RRT* and A* in terms of comprehensive performance, especially in planning time and path smoothness, while maintaining high competitiveness in path length and safety.

IV. DISCUSSION

Research in UAV path planning has witnessed continuous advancements, with scholars proposing diverse algorithms to address inherent challenges. While the classical APF method offers computational efficiency and real-time responsiveness [6], it suffers from fundamental limitations including target unreachability, local minima, and local path oscillation. To overcome these issues, this study proposes an Improved IAPF method that integrates environmental perception with deep learning mechanisms. By introducing three optimization strategies—target distance-weighted repulsion fields, local exploration factors, and directional weighting coefficients—the proposed method significantly enhances path planning performance.

For the target unreachability problem, although Khatib's foundational APF framework [6] laid the groundwork for subsequent studies, it failed to resolve terminal convergence failures. Our solution incorporates a target distance-weighted function to dynamically adjust the balance between attraction and repulsion forces, ensuring enhanced gravitational dominance as the UAV approaches the target. This approach aligns with the methodology of modifying gravitational field thresholds and adjustment factors [24], both aiming to reinforce gravitational influence in terminal phases.

Regarding local minima, existing studies have explored various solutions such as dynamic adjustment mechanisms [17], adaptive inertia weights [18], and repulsion field modifications [19]. The local exploration factor introduced in this work shares conceptual similarities with these strategies, employing stochastic perturbations to escape local traps—a principle analogous to [22] auxiliary obstacle-avoidance force mechanism.

To mitigate local path oscillation, prior research has adopted memory-based resultant forces [25], deep reinforcement learning integration [26], and collision risk assessment [27]. Our directional weighting coefficient strategy similarly focuses on reducing oscillatory behavior near obstacles by constraining abrupt heading changes, mirroring [25] philosophy of leveraging historical motion data for path smoothing.

The IAPF method also demonstrates notable achievements in path optimization. By assigning threat-specific distances to distinct obstacle types and implementing polynomial fitting for trajectory smoothing, our approach enhances flight safety while minimizing path length. This aligns with [26] convergent approach combining APF with deep reinforcement learning, prioritizing safety without compromising path efficiency.

In conclusion, the IAPF method synthesizes multiple optimization strategies to address critical shortcomings of classical APF in UAV path planning. Compared to existing literature, it demonstrates marked improvements in resolving target unreachability, local minima, and oscillations, providing a robust and efficient solution for autonomous navigation in complex environments. However, this study acknowledges that IAPF still has certain limitations when facing concave obstacles (such as ''U''-shaped canyons). Future work will focus on exploring hybrid architectures combining IAPF with global planners like RRT and dynamic extensions integrating reinforcement learning (RL), as well as verifying its scalability in larger spaces with more obstacles, to further enhance its generalization ability and practicality in complex environments.

A. CONCLUSION

To address the path planning problem of UAVs in a three-dimensional environment, this paper proposes an improved artificial potential field UAV path planning method. The algorithm has been improved respectively for the three problems of the traditional artificial potential field method: For the target unreachability problem, the repulsive potential field function is improved by introducing a target point distance - weighted function; for the local minima problem, a local exploration factor is introduced to guide the UAV to jump out of the local minima point; for the local path oscillation problem, the situation is improved by introducing a directional weighting factor and a forward - looking decision - making mechanism. Finally, the path is optimized by B - spline fitting to generate a smoother path.

REFERENCES

[1] X. Wang, Y. Wang, X. Su, L. Wang, C. Lu, H. Peng, and J. Liu, ''Deep reinforcement learning-based air combat maneuver decision-making: Literature review, implementation tutorial and future direction,'' *Artif. Intell. Rev.*, vol. 57, no. 1, pp. 21–64, Jan. 2024, doi: 10.1007/s10462-023-10620-2.

[2] C. Zhang and J. M. Kovacs, ''The application of small unmanned aerial systems for precision agriculture: A review,'' *Precis. Agricult.*, vol. 13, no. 6, pp. 693–712, Dec. 2012, doi: 10.1007/s11119-012-9274-5.

[3] X. Wang, H. Wang, H. Zhang, M. Wang, L. Wang, K. Cui, C. Lu, and Y. Ding, ''A mini review on UAV mission planning,'' *J. Ind. Manage. Optim.*, vol. 19, no. 5, pp. 3362–3382, May 2023, doi: 10.3934/jimo.2022089.

[4] P. Hart, N. Nilsson, and B. Raphael, ''A formal basis for the heuristic determination of minimum cost paths,'' *IEEE Trans. Syst. Sci. Cybern.*, vol. SSC-4, no. 2, pp. 100–107, Jul. 1968, doi: 10.1109/TSSC.1968.300136.

[5] E. W. Dijkstra, ''A note on two problems in connexion with graphs,'' in *Edsger Wybe Dijkstra: His Life, Work, and Legacy*, 1st ed., Cham, Switzerland: Springer, 2022, pp. 287–290, doi: 10.1145/3544585.3544600.

[6] O. Khatib, ''Real-time obstacle avoidance for manipulators and mobile robots,'' *Int. J. Robot. Res.*, vol. 5, no. 1, pp. 90–98, Mar. 1986, doi: 10.1177/027836498600500106.

[7] S. LaValle, ''Rapidly-exploring random trees: A new tool for path planning,'' Iowa State Univ., Ames, IA, USA, Res. Rep. 9811, 1998.

[8] S. Karaman, M. R. Walter, A. Perez, E. Frazzoli, and S. Teller, ''Anytime motion planning using the RRT*,'' in *Proc. IEEE Int. Conf. Robot. Autom.*, Shanghai, China, May 2011, pp. 1478–1483, doi: 10.1109/ICRA.2011.5980479.

[9] J. Kennedy and R. C. Eberhart, ''Particle swarm optimization,'' in *Proc. IEEE Int. Conf. Neural Netw.*, vol. 4, Perth, WA, Australia, Nov. 2002, pp. 1942–1948, doi: 10.1109/ICNN.1995.488968.

[10] M. Dorigo, V. Maniezzo, and A. Colorni, ''Ant system: Optimization by a colony of cooperating agents,'' *IEEE Trans. Syst., Man, Cybern., Part B (Cybernetics)*, vol. 26, no. 1, pp. 29–41, Feb. 1996, doi: 10.1109/3477.484436.

[11] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, ''Optimization by simulated annealing,'' *Science*, vol. 220, no. 4598, pp. 671–680, May 1983, doi: 10.1126/science.220.4598.671.

[12] W. Li, X. Qin, and G. Luo, ''Online trajectory planning for UAVs based on barrier-free convex regions,'' *J. Syst. Sci. Math. Sci.*, vol. 41, no. 6, pp. 1493–1506, Jun. 2021.

[13] E. Ivanjko, D. K. Nečoska, M. Gregurić, M. Vujić, G. Jurković, and S. Mandžuka, ''Ramp metering control based on the Q-learning algorithm,'' *Cybern. Inf. Technol.*, vol. 15, no. 5, pp. 88–97, Apr. 2015, doi: 10.1515/cait-2015-0019.

[14] X. Zhang, S. Xia, T. Zhang, and X. Li, ''Hybrid FWPS cooperation algorithm based unmanned aerial vehicle constrained path planning,'' *Aerosp. Sci. Technol.*, vol. 118, Nov. 2021, Art. no. 107004, doi: 10.1016/j.ast.2021.107004.

[15] G. Airlangga, R. Sukwadi, W. W. Basuki, L. F. Sugianto, O. I. A. Nugroho, Y. Kristian, and R. Rahmananta, ''Adaptive path planning for multi-UAV systems in dynamic 3D environments: A multi-objective framework,'' *Designs*, vol. 8, no. 6, p. 136, Dec. 2024, doi: 10.3390/designs8060136.

[16] M. Zhao, X. Li, Y. Lu, H. Wang, and S. Ning, ''A path-planning algorithm for autonomous vehicles based on traffic stability criteria: The AS-IAPF algorithm,'' *Mech. Sci.*, vol. 15, no. 2, pp. 613–631, Nov. 2024, doi: 10.5194/ms-15-613-2024.

[17] S. Min and H. Nam, ''ACACT: Adaptive collision avoidance algorithm based on estimated collision time for swarm UAVs,'' *IEEE Access*, vol. 11, pp. 120179–120191, 2023, doi: 10.1109/ACCESS.2023.3327928.

[18] H. Sang, Y. You, X. Sun, Y. Zhou, and F. Liu, ''The hybrid path planning algorithm based on improved A* and artificial potential field for unmanned surface vehicle formations,'' *Ocean Eng.*, vol. 223, Mar. 2021, Art. no. 108709, doi: 10.1016/j.oceaneng.2021.108709.

[19] C. Zhu, W. Zhao, Q. Li, P. Li, and Q. Da, "Network embedding-based anomalous density searching for multi-group collaborative fraudsters detection in social media," *Comput., Mater. Continua*, vol. 60, no. 1, pp. 317–333, 2019, doi: 10.32604/cmc.2019.05677.

[20] X. Lin, Y.-J. Yu, S.-Z. Chen, and Y.-Y. Shi, "An improved APF method for complex and dynamic obstacles' avoidance," *Unmanned Syst.*, vol. 11, no. 2, pp. 175–189, Apr. 2023, doi: 10.1142/s2301385023410054.

[21] Z. Xu, J. Hu, Y. Ma, M. Wang, and C. Zhao, "Research on UAV collision avoidance path planning algorithm," *J. Northwestern Polytech. Univ.*, vol. 37, no. 1, pp. 100–106, Feb. 2019.

[22] Q. Lv, G. Hao, Z. Huang, B. Li, D. Fu, H. Zhao, W. Chen, and S. Chen, "Localized path planning for mobile robots based on a subarea-artificial potential field model," *Sensors*, vol. 24, no. 11, p. 3604, Jun. 2024, doi: 10.3390/s24113604.

[23] F. Yang and H. Su, "Research on improving artificial potential field for drone obstacle avoidance," in *Proc. 3rd Int. Conf. Artif. Intell. Comput. Inf. Technol. (AICIT)*, Sep. 2024, pp. 1–4, doi: 10.1109/aicit62434.2024.10730002.

[24] L. Zhang, J. Mou, P. Chen, and M. Li, "Path planning for autonomous ships: A hybrid approach based on improved APF and modified VO methods," *J. Mar. Sci. Eng.*, vol. 9, no. 7, p. 761, Jul. 2021, doi: 10.3390/jmse9070761.

[25] F. Meng, X. Sun, J. Zhu, B. Mei, and P. Zheng, "Research on ship path planning based on bidirectional A*-APF algorithm," in *Proc. 4th Int. Conf. Consum. Electron. Comput. Eng. (ICCECE)*, Guangzhou, China, 2024, pp. 460–466, doi: 10.1109/ICCECE61317.2024.10504166.

[26] Z. Xi, H. Han, J. Cheng, and M. Lv, "Reducing oscillations for obstacle avoidance in a dense environment using deep reinforcement learning and time-derivative of an artificial potential field," *Drones*, vol. 8, no. 3, p. 85, Mar. 2024, doi: 10.3390/drones8030085.

[27] G. Hao, Q. Lv, Z. Huang, H. Zhao, and W. Chen, "UAV path planning based on improved artificial potential field method," *Aerospace*, vol. 10, no. 6, p. 562, Jun. 2023, doi: 10.3390/aerospace10060562.

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